The equation shows that you will need the same number of moles of sodium hydrogencarbonate. Therefore we know that we need 0.00250 mol of $NaHCO_3$.

To calculate the volume this is in, we just need to rearrange our concentration equation to obtain:

$$\text{volume (dm}^3\text{)} = \frac{\text{number of moles (mol)}}{\text{concentration (mol/dm}^3\text{)}}$$

$$\text{volume} = \frac{0.00250}{0.100} = 0.0250 \text{ dm}^3$$

The volume comes out in dm^3 because the concentration is in mol/dm^3. We can convert to cm^3 by multiplying by 1000:

volume of $NaHCO_3$ solution = 25.0 cm^3

You will need 25.0 cm^3 of the sodium hydrogencarbonate solution to neutralise the hydrochloric acid.

CHAPTER QUESTIONS

SKILLS PROBLEM SOLVING

In questions 1 to 4, take the molar volume to be 24 dm^3 (24 000 cm^3) at rtp.

8

1 Calculate the amount in moles of each of the following:

a 2.4 dm^3 of O_2 at rtp
b 480 dm^3 of He at rtp
c 100 cm^3 of CO_2 at rtp
d 1500 cm^3 of N_2 at rtp

2 Calculate the volume of each of the following gases at rtp:

a 2.0 mol H_2
b 0.10 mol SO_2
c 1.0×10^{-3} mol CO

3 a Calculate the mass of 200 cm^3 of chlorine gas (Cl_2) at rtp. (A_r: Cl = 35.5)

b Calculate the volume occupied by 0.16 g of oxygen (O_2) at rtp. (A_r: O = 16)

c If 1 dm^3 of a gas at rtp has a mass of 1.42 g, calculate the mass of 1 mole of the gas.

9

4 0.240 g of magnesium is reacted with an excess of dilute sulfuric acid. (A_r: Mg = 24)

$$Mg(s) + H_2SO_4(aq) \rightarrow MgSO_4(aq) + H_2(g)$$

Calculate the amount in moles of Mg which reacted.

Calculate the number of moles of hydrogen produced in the reaction.

Calculate the volume of hydrogen (measured at rtp) produced in the reaction.

5 Potassium nitrate decomposes when heated to produce oxygen gas:

$$2KNO_3(s) \rightarrow 2KNO_2(s) + O_2(g)$$

In an experiment 1.00 dm^3 (measured at rtp) of oxygen gas was collected.

(A_r: N = 14, O = 16, K = 39; molar volume = 24 dm^3 at rtp)

Calculate the amount in moles of oxygen gas collected.

Calculate the amount in moles of KNO_3 that reacted.

Calculate the mass of KNO_3 that reacted.

SKILLS PROBLEM SOLVING

6 Chlorine can be prepared by heating manganese(IV) oxide with an excess of concentrated hydrochloric acid. What is the maximum volume of chlorine (measured at room temperature and pressure) that could be obtained from 2.00 g of manganese(IV) oxide? (A_r: O = 16, Mn = 55; molar volume = 24 000 cm^3 at rtp)

$$MnO_2(s) + 4HCl(aq) \rightarrow MnCl_2(aq) + Cl_2(g) + H_2O(l)$$

7 Some dilute sulfuric acid, H_2SO_4, had a concentration of 4.90 g/dm^3. What is its concentration in mol/dm^3? (A_r: H = 1, O = 16, S = 32)

8 What is the concentration in g/dm^3 of potassium hydroxide, KOH, solution with a concentration of 0.200 mol/dm^3? (A_r: H = 1, O = 16, K = 39)

9 Calculate the amount in moles in each of the following:

a 25.0 cm^3 of 0.100 mol/dm^3 NaCl(aq)

b 200 cm^3 of 0.200 mol/dm^3 H_2SO_4(aq)

c 75.0 cm^3 of 0.150 mol/dm^3 HCl(aq)

d 22.4 cm^3 of 0.280 mol/dm^3 HNO_3(aq)

10 Calculate the concentration in mol/dm^3 of each of the following solutions:

a 2.00 dm^3 of sodium hydroxide solution containing 0.100 mol sodium hydroxide

b 25.0 cm^3 of sulfuric acid containing 0.0200 mol sulfuric acid

c 27.8 cm^3 of hydrochloric acid containing 0.00150 mol hydrochloric acid

11 Calculate the volume in cm^3 of each of the following solutions:

a the volume of 0.100 mol/dm^3 H_2SO_4(aq) that contains 0.500 mol

b the volume of 0.0200 mol/dm^3 NaOH(aq) that contains 0.00500 mol

c the volume of 0.500 mol/dm^3 $MgCl_2$(aq) that contains 0.0200 mol

12 When barium chloride solution is added to copper(II) sulfate solution a precipitate of barium sulfate ($BaSO_4$) is formed.

$$BaCl_2(aq) + CuSO_4(aq) \rightarrow BaSO_4(s) + CuCl_2(aq)$$

Excess barium chloride solution is added to 20.0 cm^3 of copper(II) sulfate solution of concentration 0.100 mol/dm^3. (A_r: O = 16, S = 32, Ba = 137)

Calculate the number of moles of copper(II) sulfate.

Calculate the number of moles of barium sulfate formed.

Calculate the mass of barium sulfate formed.

13 Calcium carbonate reacts with hydrochloric acid:

$$CaCO_3(s) + 2HCl(aq) \rightarrow CaCl_2(aq) + H_2O(l) + CO_2(g)$$

Calcium carbonate is added to 25.0 cm^3 of 2.00 mol/dm^3 hydrochloric acid. (A_r: C = 12, O = 16, Ca = 40)

Calculate the amount in moles of hydrochloric acid.

Calculate the amount in moles of $CaCO_3$ that reacts with the acid.

Calculate the mass of $CaCO_3$ that reacts with the acid.

Calculate the volume of CO_2 (measured at rtp) produced.

(Molar volume at rtp is 24 000 cm^3.)

SKILLS PROBLEM SOLVING



14 Hydrogen peroxide solution decomposes to form oxygen gas:

$$2H_2O_2(aq) \rightarrow 2H_2O(l) + O_2(g)$$

Calculate the volume of oxygen gas (measured at rtp) produced when 30.0 cm^3 of 0.0200 mol/dm^3 hydrogen peroxide decomposes to form oxygen gas. (Molar volume at rtp is 24 000 cm^3.)

15 **a** Calculate the volume of 0.200 mol/dm^3 sulfuric acid needed to neutralise 25.0 cm^3 of 0.400 mol/dm^3 sodium hydroxide solution.

$$2NaOH(aq) + H_2SO_4(aq) \rightarrow Na_2SO_4(aq) + 2H_2O(l)$$

b Calculate the minimum volume of 2.00 mol/dm^3 hydrochloric acid needed to react with 10.0 g of calcium carbonate. (A_r: C = 12, O = 16, Ca = 40)

$$CaCO_3(s) + 2HCl(aq) \rightarrow CaCl_2(aq) + H_2O(l) + CO_2(g)$$

16 In each of these questions concerning titrations, calculate the unknown concentration in mol/dm^3.

a 25.0 cm^3 of 0.100 mol/dm^3 sodium hydroxide was neutralised by 20.0 cm^3 of dilute nitric acid of unknown concentration.

$$NaOH(aq) + HNO_3(aq) \rightarrow NaNO_3(aq) + H_2O(l)$$

b 25.0 cm^3 of sodium carbonate solution of unknown concentration was neutralised by 30.0 cm^3 of 0.100 mol/dm^3 nitric acid.

$$Na_2CO_3(aq) + 2HNO_3(aq) \rightarrow 2NaNO_3(aq) + CO_2(g) + H_2O(l)$$

c 25.0 cm^3 of 0.250 mol/dm^3 potassium carbonate solution was neutralised by 12.5 cm^3 of ethanoic acid of unknown concentration.

$$2CH_3COOH(aq) + K_2CO_3(aq) \rightarrow 2CH_3COOK(aq) + CO_2(g) + H_2O(l)$$

17 Limewater is calcium hydroxide solution. In an experiment to find the concentration of calcium hydroxide in limewater, 25.0 cm^3 of limewater needed 18.8 cm^3 of 0.0400 mol/dm^3 hydrochloric acid to neutralise it.

$$Ca(OH)_2(aq) + 2HCl(aq) \rightarrow CaCl_2(aq) + 2H_2O(l)$$

Calculate the concentration of the calcium hydroxide in:

a mol/dm^3 **b** g/dm^3

(A_r: H = 1, O = 16, Ca = 40)

18 Sodium carbonate solution reacts with nitric acid according to the following equation:

$$Na_2CO_3(aq) + 2HNO_3(aq) \rightarrow 2NaNO_3(aq) + CO_2(g) + H_2O(l)$$

In each of the following questions work out which reagent is in excess and calculate the volume of CO_2 produced at rtp. (Molar volume at rtp is 24000 cm^3)

a 0.1 mol Na_2CO_3 is reacted with 0.1 mol HNO_3.

b 20.0 cm^3 of 0.100 mol/dm^3 Na_2CO_3 is reacted with 0.0200 mol HNO_3.

c 25.0 cm^3 of 0.300 mol/dm^3 Na_2CO_3 is reacted with 20.0 cm^3 of 0.400 mol/dm^3 HNO_3.

END OF CHEMISTRY ONLY